

Aditya Kumar Gupta

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SUMMARY

Final-year Information Science student with strong proficiency in Python, SQL, and AI/ML fundamentals. Experienced in building end-to-end AI systems, including an LLM-based self-healing validation system and a crop yield prediction system. Skilled in machine learning and backend development using FastAPI and REST APIs. Hands-on experience in designing scalable pipelines, API integrations, and improving system reliability. Seeking entry-level roles in AI/ML, Data Science, or Backend development.

WORK EXPERIENCE

Android Developer Intern — MindMatrix.io Feb 2026 – Present

- Developed AI-powered applications integrating Generative AI APIs for real-time interactions.
- Designed scalable backend APIs enabling seamless communication between applications and AI services.
- Optimized system performance and response handling, improving reliability and reducing intervention.

PROJECTS

AI Self-Healing Validation System—Python, FastAPI, LLM, LangGraph [GitHub](#) — [Demo](#)

- Built an AI-driven system to detect and correct incorrect outputs using LLM-based validation pipelines.
- Designed a multi-step workflow (validation → correction → re-evaluation) to improve response reliability.
- Integrated LLM APIs with backend services using FastAPI for automated issue detection and resolution.
- Reduced manual intervention by ~30% by implementing iterative self-healing mechanisms.

Smart Crop Yield Prediction System —Python, FastAPI, Scikit-Learn [GitHub](#) — [Demo](#)

- Built a FastAPI-based REST API to process multi-variate agricultural inputs for crop yield prediction.
- Developed a Random Forest model trained on 43,000+ data points across crops and regions.
- Designed preprocessing pipelines using ColumnTransformer with scaling, encoding, and data handling.
- Improved decision support by delivering data-driven insights through an interactive dashboard interface.

EDUCATION

2022 - present	B.E at Sir M. Visvesvaraya Institute of Technology, Bengaluru	(GPA: 8.2/10.0)
2019 - 2021	Intermediate(CBSE) at Holy Mission High School, Samastipur	(Percent: 70.8)
2018 - 2019	Matriculation(CBSE) at St. Xaviers High School, Samastipur	(Percent: 77)

TECHNICAL SKILLS

Languages:	Python, JavaScript, SQL (MySQL), HTML, CSS
Backend & APIs:	FastAPI, RESTful APIs, Uvicorn, Pydantic, JWT Auth
Machine Learning & Data:	Scikit-learn (Random Forest), Pandas, NumPy, Feature Engineering, Model Evaluation (RMSE, R ²)
AI & LLMs:	LLM APIs (Groq), LangGraph, AI Agent Systems
Data Visualization & Core Concepts:	Chart.js, Streamlit, OOP, DBMS, SQLite, OS, CN, DSA

CERTIFICATIONS / VIRTUAL EXPERIENCE PROGRAMS

JPMorgan Chase & Co. – Quantitative Research Virtual Experience (Forage) Oct 2025

- Analyzed loan data to estimate default probabilities using quantitative research methods.